Markscheme

May 2018

Chemistry

Higher level

Paper 2

22 pages

Question 1

Question		Answers	Notes	Total
1.	**a**	$n(HNO_3) = 0.0500\ dm^3 \times 0.120\ mol\ dm^{-3} = 0.00600 / 6.00 \times 10^{-3}$ mol ✓		**1**
1.	**b**	$3HNO_3\ (aq) + Al(OH)_3\ (s) \rightarrow Al(NO_3)_3\ (aq) + 3H_2O\ (l)$ ✓	*Accept ionic equation.*	**1**
1.	**c**	$n(HNO_3) = n(NaOH) = 0.02240\ dm^3 \times 0.1050\ mol\ dm^{-3} = 0.002352 / 2.352 \times 10^{-3}$ mol ✓		**1**
1.	**d**	$n(HNO_3)$ reacted = 0.00600 – 0.002352 = 0.003648 / 3.648×10^{-3} mol ✓		**1**
1.	**e**	$n(Al(OH)_3) = (1/3) \times n(HNO_3) = (1/3) \times 0.003648 = 1.216 \times 10^{-3}$ mol ✓ $m(Al(OH)_3) = 1.216 \times 10^{-3} \times 78.00 = 0.0948$ g ✓	*Award [2] for correct answer.* *Award [1 max] for 0.0948.*	**2**
1.	**f**	% $Al(OH)_3$ = (0.0948 / 1.62) × 100 = 5.85% ✓	*Answer must be to 3 s.f.*	**1**
1.	**g**	to reduce random errors OR to increase precision ✓	*Accept "to check for systematic errors".*	**1**

Question 2

Question		Answers	Notes	Total
2.	**a**	both axes correctly labelled ✓ correct shape of curve starting at origin ✓ $E_{a(catalyst)} < E_{a(without\ catalyst)}$ on x-axis ✓	*M1: Accept "speed" for x-axis.* *M2: Curve must not touch x-axis.* *M3: Ignore shading.*	**3**
2.	**b i**	curve starting from origin with steeper gradient AND reaching same maximum volume ✓		**1**
2.	**b ii**	rate decreases OR slower reaction ✓ propanoic acid partially dissociated/ionized in solution/water OR lower $[H^+]$ ✓	*Accept "weak acid" or "higher pH".*	**2**
2.	**c**	pH converts wide range of $[H^+]$ into simple log scale/numbers ✓ OR avoids exponential notation OR converts small numbers into 0–14 range		**1**
2.	**d i**	A: CH_3CH_2COOH AND $CH_3CH_2COO^-$ ✓ B: $CH_3CH_2COO^-$ ions ✓	*Penalise "instead of" once.*	**2**
2.	**d ii**	$K_a = 1.34 \times 10^{-5} = [H^+]^2/0.10$ ✓ $[H^+] = 1.158 \times 10^{-3}$; pH = 2.94 ✓		**2**
2.	**d iii**	$CH_3CH_2COO^-$ (aq) + H_2O (l) ■ CH_3CH_2COOH (aq) + OH^- (aq) ✓		**1**
2.	**d iv**	less than 7 ✓		**1**
2.	**e i**	SO_2 (g) + H_2O (l) $\rightarrow$ H_2SO_3 (aq) ✓		**1**
2.	**e ii**	H_2SO_3 (aq) + $ZnCO_3$ (s) $\rightarrow$ $ZnSO_3$ (aq) + CO_2 (g) + H_2O (l) ✓		**1**

Question 3

Question		Answers	Notes	Total
3.	**a i**	4 levels showing convergence at higher energy ✓		**1**
3.	**a ii**	arrows (pointing down) from n = 3 to n = 2 AND n = 4 to n = 2 ✓		**1**
3.	**a iii**	IE = hν = $6.63 \times 10^{-34} \times 3.28 \times 10^{15} = 2.17 \times 10^{-18}$ J ✓		**1**
3.	**a iv**	$\lambda = c/\nu = 3.00 \times 10^{8} / 3.28 \times 10^{15} = 9.15 \times 10^{-8}$ m ✓		**1**
3.	**b i**	same shielding AND nuclear charge/Z_{eff} increases ✓		**1**
3.	**b ii**	K^{+} 19 protons AND Cl^{-} 17 protons OR K^{+} has more protons ✓ same number of electrons/isoelectronic thus pulled closer together ✓		**2**
3.	**c i**	4s: ↑↓ 3d: ↑↓ ↑↓ ↑↓ ↑↓ ↑ *Cu is [Ar] $3d^{10}4s^{1}$*		**1**
3.	**c ii**	Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$ ✓ Cathode: $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$ ✓	*Accept ■ for →. Award [1 max] if at wrong electrodes.*	**2**
3.	**c iii**	external circuit AND from anode to cathode (positive to negative) ✓		**1**
3.	**c iv**	no change in colour ✓		**1**
3.	**c v**	oxygen / O_2 ✓		**1**
3.	**d**	Transition metals: d and s orbitals close in energy / ionization energies increase gradually ✓ Alkali metals: second electron removed from lower energy level / large jump in IE ✓		**2**

Question 4

Question		Answers	Notes	Total
4.	**a**	$ClO_3^-(aq) + 6H^+(aq) + 6Fe^{2+}(aq) \rightarrow Cl^-(aq) + 3H_2O(l) + 6Fe^{3+}(aq)$ ✓	*Accept* ■ *for* →.	**1**
4.	**b**	n = 6 ✓ $E^{\ast} = -(-412 \times 10^3) / (6 \times 96500) = 0.711\ V$ ✓	*Award [2] for correct answer.*	**2**
4.	**c**	$E^{\ast}(ClO_3^-/Cl^-) = 0.711 + 0.77 = 1.48\ V$ ✓		**1**

Question 5

Question		Answers	Notes	Total
5.	**a**	bonds broken: 4(C–H) + 2(H–O) = 4(414) + 2(463) = 2582 kJ ✓ bonds made: 3(H–H) + C≡O = 3(436) + 1077 = 2385 kJ ✓ ΔH = 2582 – 2385 = +197 kJ ✓	*Award [3] correct answer.* *Award [2 max] for –197 kJ.*	**3**
5.	**b i**	$\Delta H^{\ominus}_f$ for any element = 0 by definition ✓		**1**
5.	**b ii**	$\Delta H^{\ominus}$ = –110.5 + 0 – [–74.6 + (–241.8)] = +205.9 kJ ✓		**1**
5.	**b iii**	bond enthalpies are averaged values over similar compounds ✓		**1**
5.	**c**	$\Delta S^{\ominus}$ = 198 + 3(131) – (186 + 189) = +216 J K^{-1} ✓		**1**
5.	**d**	$\Delta G^{\ominus}$ = 205.9 – 298 × (216/1000) = +141.5 kJ ✓	*Accept answer consistent with (b)(ii).*	**1**
5.	**e**	T = $\Delta H^{\ominus}/\Delta S^{\ominus}$ = 205900/216 = 953 K ✓		**1**

Question 6

Question		Answers	Notes	Total
6.	**a**	Q: non-equilibrium concentrations AND K_c: equilibrium concentrations ✓		**1**
6.	**b**	$Q = [SO_3]^2/[SO_2]^2[O_2] = (2.00)^2/(2.00)^2(1.00) = 1.00$ ✓ reverse reaction favoured AND $Q > K_c$ / $1.00 > 0.500$ ✓	*Do not award M2 without comparison.*	**2**
6.	**c i**	$[N_2O_2]$ decreases AND exothermic so reverse reaction favoured ✓		**1**
6.	**c ii**	$K_c = [N_2O_2]/[NO]^2$ ✓ rate $= k_2[N_2O_2][O_2] = k[NO]^2[O_2]$ ✓	*Award [2] for correct expression.*	**2**
6.	**d**	$T_1 = 293$ K, $T_2 = 303$ K ✓ $\ln(3) = E_a/8.314 \times (1/293 - 1/303)$; $E_a = 75.8$ kJ mol^{-1} ✓	*Award [2] for correct answer.*	**2**

Question 7

Question		Answers	Notes	Total
7.	a i	polar bonds between H and group 16 element ✓ non-linear/bent shape due to lone pairs OR dipoles do not cancel ✓	*M2: Do not accept "has a dipole" alone.*	2
7.	a ii	number of electrons increases ✓ London/dispersion forces increase ✓		2
7.	b	Electron domain geometry: tetrahedral ✓ Molecular geometry: bent/V-shaped/angular ✓ *Note: PH_2^- same geometry as NH_2^-*	*Both marks from correct diagram.*	2
7.	c i	Structure I: O(1)=0, O(2)=0 ✓ Structure II: O(1)=–1, O(2)=+1 ✓		2
7.	c ii	structure I AND no formal charges / no charge transfer ✓		1
7.	d	O_3 bond order 1.5 AND O_2 bond order 2 / O_3 weaker bond ✓ O_3 requires longer wavelength ✓		2
7.	e	CO_2 non-polar / weak London forces ✓ SiO_2 giant covalent/network structure ✓		2

Turn over

Question 8

Question		**Answers**	**Notes**	**Total**
8.	**a**	Physical: equal C–C bond lengths OR all C–C bond order 1.5 ✓ Chemical: undergoes substitution more readily / does not decolourise bromine water ✓		**2**
8.	**b i**	$3CH_3CH_2CH_2CH_2OH + Cr_2O_7^{2-} + 8H^+ \rightarrow 3CH_3CH_2CH_2CHO + 2Cr^{3+} + 7H_2O$ ✓ correct reactants/products ✓ balanced ✓		**2**
8.	**b ii**	Aldehyde: distillation (removed as formed) ✓ Carboxylic acid: reflux ✓		**2**
8.	**c i**	$C_8H_8O_2$ ✓		**1**
8.	**c ii**	A: C–H (alkanes/arenes) AND B: C=O (carbonyl/ester) ✓		**1**
8.	**c iii**	Any two of: $C_6H_5COOCH_3$ ✓ OR $CH_3COOC_6H_5$ ✓ OR $HCOOCH_2C_6H_5$ ✓	*Award [1 max] for two aliphatic esters.*	**2**
8.	**c iv**	$C_6H_5COOCH_3$ signal at ~4 ppm due to alkyl group on ester ✓		**1**

Question 9

Question		**Answers**	**Notes**	**Total**
9.	**a i**	$CH_3CH_2COCH_2CH_2CH_3$ (hexan-3-one) ✓ $CH_3CH_2CH_2COCH_2CH_3$ (hexan-2-one alternative) ✓	*Accept condensed formulas.*	**2**
9.	**a ii**	A: peaks at 43 or 71 due to symmetrical ketone fragments ✓ B: peaks at 29 or 57 due to asymmetric fragments ✓	*Penalise missing + once.*	**2**
9.	**b i**	heterolytic / heterolysis ✓		**1**
9.	**b ii**	polar protic ✓		**1**
9.	**b iii**	carbocation / $R_1R_2R_3C^+$ ✓ Shape: triangular/trigonal planar ✓		**2**
9.	**b iv**	~50% each OR equal percentages ✓ nucleophile attacks from either side of planar carbocation ✓	*Accept "racemic mixture".*	**2**
9.	**c**	Stage 1: $C_6H_5NO_2 + 3Sn + 7H^+ \rightarrow C_6H_5NH_3^+ + 3Sn^{2+} + 2H_2O$ ✓ Stage 2: $C_6H_5NH_3^+ + OH^- \rightarrow C_6H_5NH_2 + H_2O$ ✓		**2**

Turn over